LONGEVITY 04

Metformin

Biguanide blood sugar medicine taken by mouth



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Biguanide	Type 2 diabetes, aging biology	Large trials; FDA approved	FDA approved (diabetes) · aging use off-label

STUDIED FOR

- **Blood sugar** — HUMAN
- **Exercise gains** — HUMAN
- **Survival** — HUMAN
- **Aging biology** — HUMAN + ANIMAL

No aging trial result yet. The TAME aging trial was proposed in 2016; no registry record was found. Lifespan gains are from model systems.

HOW IT WORKS IN THE BODY

Metformin ▶ Liver glucose ▶ Glucose uptake ▶ Lower blood sugar

- **Decreases liver glucose output** and gut glucose absorption (label)
- **Improves insulin sensitivity** by raising glucose uptake and use (label)
- **Healthspan and lifespan** gains are from animal and model systems
- **In people,** an effect on aging is not established

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approved for type 2 diabetes since 1995
- Aging: small trials, 14 to 94 older adults
- Survival studies in diabetes disagree

NOT YET ESTABLISHED

- Effect on aging in people
- Heart and artery risk reduction

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
About 6.2 hours in plasma; 17.6 hours in blood	By mouth (tablets)
PHARMACOKINETICS	**SOLD AS**
Not metabolized by the liver; excreted unchanged in urine	Prescription tablets (Glucophage, generics)
MONITORED IN TRIALS	**SPORT (WADA)**
Kidney function (eGFR), vitamin B12, blood sugar	Not on the 2026 list

REGULATORY

FDA	Approved 1995, type 2 diabetes; aging off-label
FDA	Boxed warning: lactic acidosis
WADA	Not named on the list

REFERENCES

1. FDA, metformin hydrochloride tablets label (DailyMed), Oct 2026
2. Cell Metab, Metformin as a tool to target aging (TAME), Jun 2016
3. WADA, 2026 Prohibited List

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